EXAMPLE 3.25

The coil of a moving coil galvanometer has 5 turns and each turn has an effective area of 2×10^{-2} m^2. It is suspended in a magnetic field whose strength is 4×10^{-2} Wb m^{-2}. If the torsional constant K of the suspension fibre is 4×10^{-9} N m deg^{-1}.

(a) Find its current sensitivity in division per micro - ampere.

(b) Calculate the voltage sensitivity of the galvanometer for it to have full scale deflection of 50 divisions for 25 mV.

(c) Compute the resistance of the galvanometer.

Solution

$N = 5$ turns

$A = 2 \times 10^{-2}$ m^2

$B = 4 \times 10^{-2}$ Wbm^{-2}

$K = 4 \times 10^{-9}$ N m deg^{-1}

(a) Current sensitivity

$$I_S = \frac{NAB}{K} = \frac{5 \times 2 \times 10^{-2} \times 4 \times 10^{-2}}{4 \times 10^{-9}}$$

$= 10^6$ divisions per ampere

$1\mu A = 1 \text{micro ampere} = 10^{-6} \text{ampere}$

Therefore,

$$I_S = 10^6 \frac{\text{div}}{\text{A}} = 1 \frac{\text{div}}{10^{-6}\text{A}} = 1 \frac{\text{div}}{\mu\text{A}}$$

$$I_S = 1 \text{ div} (\mu\text{A})^{-1}$$

(b) Voltage sensitivity

$$V_S = \frac{\theta}{V} = \frac{50 \text{ div}}{25 \text{ mV}} = 2 \times 10^3 \text{ div V}^{-1}$$

(c) The resistance of the galvanometer is

$$R_g = \frac{I_S}{V_S} = \frac{10^6 \frac{\text{div}}{\text{A}}}{2 \times 10^3 \frac{\text{div}}{\text{V}}} = 0.5 \times 10^3 \frac{\text{V}}{\text{A}} = 0.5 \text{ k}\Omega$$

EXAMPLE 3.26

The resistance of a moving coil galvanometer is made twice its original value in order to increase current sensitivity by 50%. Find the percentage change in voltage sensitivity.

Solution

Voltage sensitivity is $V_S = \frac{I_S}{R_g}$

When the resistance is doubled, then new resistance is $R_g' = 2R_g$

Increase in current sensitivity is

$$I_S' = \left(1 + \frac{50}{100}\right) I_S = \frac{3}{2} I_S$$

The new voltage sensitivity is

$$V_S' = \frac{\frac{3}{2} I_S}{2R_g} = \frac{3}{4} V_S$$

Hence the voltage sensitivity decreases. The percentage decrease in voltage sensitivity is

$$\frac{V_S - V_S'}{V_S} \times 100\% = 25\%$$

Conversion of galvanometer into ammeter and voltmeter

A galvanometer is very sensitive instrument to detect the current. It can be easily converted into ammeter and voltmeter.

Galvanometer to an Ammeter

Ammeter is an instrument used to measure current flowing in the electrical circuit. The ammeter must offer low resistance such that it will not change the current passing through it. So ammeter is connected in series to measure the circuit current.

12th Physics_Vol-1_Unit 3_English Medium.indd 183 23-12-2022 15:03:54